Transformation of Random Variables

CSE400_L11.pdf
PDF

Generate a lecture scribe intended to serve as accurate and exam-oriented reference material for revision. Use only the provided lecture slides (attached pdf) as the authoritative source.
The scribe must faithfully reconstruct what was taught in the lecture so that a student studying only this document can reliably recall definitions, notation, assumptions, results, derivations, and example solutions.
Include only material present in the provided context, and do not introduce new content.
The scribe must include, wherever present in the context:
• Definitions and notation
• Assumptions and required conditions
• Statements of results or theorems
• Proofs or derivations presented in lecture, written step by step
• Worked examples with intermediate steps exactly as shown
• All solutions provided in the lecture slides or PDF
• Solution steps exactly in the same logical order and form as presented in the context
• Clear logical flow between concepts and sections
If the lecture contains a fully solved example, including the derivation for the transformation involving the random variables $Z=X+Y$, the complete solution must be included in the lecture scribe with all steps exactly as presented in the lecture material, without modifying, skipping, or adding steps.
Follow the lecture's structure and scope. Maintain a formal academic tone suitable for exam preparation.
Organize the scribe clearly using logical sections and headings so that it can be directly used for revision.
Do not add intuition, extra explanations, simplifications, new examples, or external material not present in the context.

Do not invent missing steps or introduce content beyond the lecture or textbook material.
The output must strictly prioritize correctness, faithfulness to lecture content, and structured presentation for exam preparation.

CSE400 — Lecture 11

Transformation of Random Variables

(Exam-Oriented Lecture Scribe for Revision)

Course: CSE400 — Fundamentals of Probability in Computing
Lecture Topic: Transformation of Random Variables
Instructor: Dhaval Patel, Associate Professor, SEAS, Ahmedabad University
Lecture Date: February 10, 2026

This scribe reconstructs the lecture content strictly from the lecture slides and preserves the structure, results, and solution steps presented.

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1. Lecture Outline

The lecture covers:

1. **Transformation of Random Variables**
   - Learning techniques to transform random variables.
2. **Function of Two Random Variables**
   - Joint transformations and derived distributions.
3. **Illustrative Example**
   - Detailed derivation for the case:

$$Z = X + Y$$

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2. Transformation of Random Variables

The lecture considers transformation of a random variable where:

- A random variable X is known.
- A new random variable Y is defined as a function of X:

$$Y = g(X)$$

The goal is to obtain the distribution of Y.

Method Used in Lecture: CDF-Based Transformation

The cumulative distribution function (CDF) of Y is derived using probability relations.

Step-by-step derivation shown:

1. Define CDF of Y:

$$F_Y(y) = P(Y \leqslant y)$$

2. Substitute transformation:

$$= P(g(X) \leqslant y)$$

3. Express event in terms of X:

$$= P(X \leqslant g^{-1}(y))$$

   (depending on monotonicity conditions of transformation).

4. Use the CDF of X:

$$= F_X(g^{-1}(y))$$

5. The PDF of Y is then obtained by differentiation:

$$f_Y(y) = \frac{d}{dy} F_Y(y)$$

This procedure uses the known PDF or CDF of X to derive that of Y.

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3. Example: Transformation of a Uniform Random Variable

An example is presented involving a **uniformly distributed random variable**.

From the slide:

- A random variable is uniformly distributed over an interval.
- A transformation is applied:

$$y = \sin(\cdot)$$

  (exact argument as shown in lecture slides).

The steps follow:

1. Start with known PDF of original variable.
2. Apply transformation relation.
3. Express transformed variable in terms of original variable.
4. Use probability relation:

$$F_Y(y) = P(Y \leqslant y)$$

5. Convert inequality into original variable's domain.
6. Use known distribution of original variable.
7. Differentiate to obtain PDF.

The lecture shows these steps symbolically, maintaining algebraic manipulation from the original variable to the transformed distribution.

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4. Function of Two Random Variables

The lecture then considers transformations involving **two random variables**.

Key setting:

- Two random variables:

$$X, Y$$

- Possibly independent.
- A new variable is defined as a function of both:

$$Z = h(X, Y)$$

The lecture focuses on determining the distribution of such derived variables using the **joint PDF**.

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5. Worked Example: Distribution of $Z = X + Y$

This is the central solved example in the lecture.

Problem Setup

A new random variable is defined:

$$Z = X + Y$$

The task is to find the distribution of Z.

Step 1: Express CDF of Z

$$F_Z(z) = P(Z \leqslant z)$$

Substituting:

$$= P(X + Y \leqslant z)$$

Step 2: Convert into joint probability

The event corresponds to all pairs (x, y) satisfying:

$$x + y \leqslant z$$

So,

$$F_Z(z) = P((X, Y) \text{ lie in this region})$$

Step 3: Use Joint PDF Integration

The lecture expresses this as integration of joint density:

$$F_Z(z) = \iint_{x+y \leqslant z} f_{X,Y}(x, y)\, dy\, dx$$

The integration is performed over the region defined by:

$$x + y \leqslant z$$

Step 4: Integration Representation Shown

The slide writes the integral in ordered form:

$$F_Z(z) = \int \int \quad f_{X,Y}(x, y)\, dy\, dx$$

with limits determined by the region $x + y \leqslant z$.

Step 5: Final Objective

Once the CDF is obtained,

$$f_Z(z) = \frac{d}{dz} F_Z(z)$$

gives the PDF of Z.

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6. Logical Flow of Lecture

The lecture proceeds in the following sequence:

1. Review transformation of a single random variable.
2. Use CDF-based method.
3. Demonstrate transformation with an example.
4. Extend method to two random variables.
5. Fully derive distribution for $Z = X + Y$.
6. Express final solution via joint PDF integration.

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7. Key Results to Remember for Exams

From this lecture:

- Transformation is done using:

$$F_Y(y) = P(Y \leqslant y)$$

- Convert event into original variable(s).
- Use known distributions.
- Differentiate to obtain PDFs.
- For two variables:

$$F_Z(z) = P(X + Y \leqslant z)$$

- Use joint PDF integration over region.

- PDF obtained by differentiation of CDF.

End of Lecture 11 Scribe

This document contains all derivations and solution steps presented in the lecture slides and is structured for direct exam revision.